

SYSTEMVERILOG Problem Set

SYSTEMVERILOG Design Problems

1. 2-Port Fair Arbiter

- File: `rr\_arbiter2`
- Description: Round-robin logic

2. Simple FIFO

- File: `fifo\_simple`
- Description: Depth=4 push/pop

3. 4-State FSM

- File: `fsm4`
- Description: IDLE, READ, WRITE, DONE

4. Handshake Protocol

- File: `handshake`
- Description: req/ack with assertions

5. Priority Arbiter

- File: `arb\_priority`
- Description: Fixed priority 4 ports

6. Round-Robin Arbiter N ports

- File: `arb\_round\_robin`
- Description: Fair arbitration

7. Dual-Port RAM

- File: `dual\_port\_ram`
- Description: Separate read/write ports

8. Parameterized FIFO

- File: `fifo\_param`
- Description: Almost full/empty flags

9. AXI-lite Slave Stub

- File: `axi\_lite\_stub`
- Description: Respond to read/write

10. Cache Line Replacement Policy

- File: `cache\_lru`
- Description: LRU logic